Topic 8 — Forests & Forest Management

Covers syllabus (click to expand)

Forests and their Types | Forest Types in India | Forest Conservation | Forest Management | Afforestation | Social Forestry (सामाजिक वानिकी) | Agroforestry (कृषि वानिकी) | Joint Forest Management (JFM) (संयुक्त वन प्रबंधन) | Forest Rights Act

Sources baked in: Champion & Seth 1968, IFA 1927, FCA 1980, NFP 1988, FRA 2006, ISFR 2023 (FSI), MoEFCC, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — ISFR data, act years, FCA vs FRA, UP district traps **Last verified:** August 2026

Current Affairs (this topic)

- Keep ISFR / FSI function traps stable; new ISFR cycle is the usual forest CA refresh.
- FCA / forest-conservation wording may appear in CA — separate from WPA PA categories.

8.1 Forests and their Types

Identity: Tree-dominated ecosystem (पारिस्थितिकी तंत्र) | Global + legal + ISFR classes

- A **forest** is a large area where trees form the dominant life form and support layered ecosystems.
- Forests regulate climate, conserve soil and water, store carbon, and provide timber and non-timber forest produce.
- **Global forest types** include tropical rainforest, tropical deciduous, temperate, boreal, Mediterranean scrub, and mangrove.
- India has no boreal forest. Its forests range from Western Ghats evergreen to Himalayan alpine scrub.
- Under the **Indian Forest Act, 1927**, forests are classed as **Reserved Forest**, **Protected Forest**, or **Unclassed Forest**.
- **Reserved Forest** has the strictest protection. Most acts are prohibited unless permitted.
- **Protected Forest** is state-regulated forest where some grazing, collection, or customary use may continue under rules.
- **Unclassed Forest** is forest land not yet formally placed in the Reserved or Protected category.
- **ISFR forest cover** means any patch of **1 hectare or more** with **10% or more** tree canopy, regardless of ownership.
- **Very Dense Forest** has canopy density **70% or more**. **Moderately Dense Forest** has **40–70%**. **Open Forest** has **10–40%**.
- **Forest cover** is not the same as **Recorded Forest Area**. Much forest cover exists outside legally notified forest land.

- **Tree cover** counts scattered trees in orchards and farms that fall below the 1-hectare forest patch rule.
- **Mangrove and littoral swamp forests** are distinct coastal types with salt-tolerant species such as sundari in the Sundarbans (सुंदरबन).

IFA class	Protection level
Reserved Forest	Strictest — hunting, grazing, felling banned unless permitted
Protected Forest	Moderate (नरम दल) — state controls but some rights allowed
Unclassed Forest	Not yet formally classified

> **Logic:** Trap — "forest cover counts only government forest land" is **false**.

8.2 Forest Types in India

Identity: Champion & Seth 1968 | 16 types | UP Terai–Vindhyan–Bundelkhand

- **Champion and Seth (1968)** is India's standard forest type classification with **16 major types** in **6 groups**.
- The **six groups** are Moist Tropical, Dry Tropical, Montane Subtropical, Montane Temperate, Sub-Alpine, and Alpine.
- **Tropical Wet Evergreen** forest occurs where rainfall exceeds about **200 cm**, mainly on the Western Ghats, Northeast hills (e.g. Arunachal, Mizoram/Khasi (खासी)–Jaintia (जयंतिया) belt), and Andaman & Nicobar — **not** Himachal (pine/temperate belts).
- These forests hold **maximum plant diversity / biomass** among common biome (जीवोम) options; equatorial rainforest is the densest high-biomass biome because of year-round high rainfall and temperature.
- **Tropical Semi-Evergreen** forest is a transition type with rainfall around **1500–2500 mm**.
- **Tropical Moist Deciduous** forest (about **100–200 cm** rain) drops leaves in the dry season; **teak** is the dominant commercial species. **Madhya Pradesh** has the maximum teak area among common state options. Other species: bamboo, sal, shisham, sandalwood, khair, arjun.
- **Red Sanders (*Pterocarpus santalinus*)** is endemic to dry deciduous hills of South India (Palkonda–Seshachalam / southern Eastern Ghats) — valued red wood; **not** a tropical rainforest species.
- **Taxus** grows naturally in the Himalaya, is Red Data–listed, and yields **taxol** (anti-cancer; older stems also cite Parkinson's).
- **Littoral and Swamp** forest includes mangroves on the Sundarbans and Odisha coast (**Sundari** is the famous Sundarbans tree).
- **Andaman & Nicobar Islands** uniquely show a combination of **mangrove + evergreen + deciduous** forests (mangrove cover ~**616–617 km²** in ISFR 2021-style figures).
- **Amazon rainforest** is popularly called the **"lungs of the Earth."**
- Oak and rhododendron are Himalayan/subtropical–temperate associates; **sandalwood** is South Indian deciduous — not a Himalayan natural associate.

- **Tropical Dry Deciduous** is India's **largest forest type** by area, covering about **38%** of forest area.
- **Tropical Thorn** forest occurs in low-rainfall zones such as **Bundelkhand** with acacia and babool at **200–800 mm**.
- **Subtropical Pine** forest is dominated by **chir pine** in the outer Himalaya between **1000–1800 m**.
- **Himalayan Moist Temperate** forest holds oak and deodar between **1500–3300 m**.
- **Sub-Alpine and Alpine scrub** occur above **3000 m** near the treeline.
- **ISFR 2023** recorded India forest cover at **21.76%** (**7,15,343 sq km**) and total green cover at **25.17%**. ISFR **2021** forest cover was about **21.71%**; forest+tree cover about **24.62%**.
- **Madhya Pradesh** has the **largest forest area** among states. **Mizoram** has the **highest percentage** forest cover. Top percentage cluster: Mizoram, Arunachal, Meghalaya, Manipur, Nagaland.
- ISFR 2021 absolute gains were led by **Andhra Pradesh, Telangana, Odisha**; the Northeast showed large losses.
- About **17** States/UTs have **>33%** forest cover; five exceed **75%** (Lakshadweep (लक्षद्वीप), Mizoram, A&N, Arunachal, Meghalaya).
- **Uttar Pradesh** (उत्तर प्रदेश) forest cover is **6.24%** of state area (ISFR 2023). **Sonbhadra** has the highest district share at **34.87%**.
- Only **Chandauli (21.63%)** and **Shravasti (20.95%)** exceed **20%** forest cover in UP.
- **Bahraich** is Terai moist deciduous by type but its forest cover is only about **10%**, below the 20% threshold.

#	Champion & Seth type	Key region
1	Tropical Wet Evergreen	Western Ghats, NE India
2	Tropical Semi-Evergreen	Transition to evergreen belt
3	Tropical Moist Deciduous	Terai, central India
4	Littoral & Swamp	Sundarbans, coastal Odisha
5	Tropical Dry Deciduous	**Largest area (~38%)**
6	Tropical Thorn	Rajasthan, Bundelkhand
7	Tropical Dry Evergreen	East coast Karnataka/AP
8	Subtropical Broadleaved Hill	Nilgiri (नीलगिरि), NE hills
9	Subtropical Pine	Chir pine — outer Himalaya
10	Subtropical Dry Evergreen	Siwalik, Jammu
11	Montane Wet Temperate	Nilgiri, eastern Himalaya
12	Himalayan Moist Temperate	Oak, deodar — Uttarakhand, HP
13	Himalayan Dry Temperate	Inner Himalaya conifers
14	Sub-Alpine	Rhododendron, fir — >3000 m
15	Moist Alpine Scrub	High Himalaya

#	Champion & Seth type	Key region
16	Dry Alpine Scrub	Cold desert Himalaya

PYQ — UPPCS Prelims 2025, Q8

As per ISFR 2023, which UP districts had forest over more than 20% of total area?

1. Bahraich 2. Chandauli 3. Shravasti

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

Show answer

Ans: C — Bahraich is below 20%.

UP region	Forest type	Key districts
Terai-Duar	Tropical Moist Deciduous	Pilibhit, Bahraich, Shravasti
Vindhyan	Tropical Dry Deciduous	Sonbhadra, Chandauli, Mirzapur (मिर्ज़ापुर)
Bundelkhand	Thorn / Dry Deciduous	Jhansi, Banda, Lalitpur

PYQ — UPPCS Prelims 2023, Q59

Consider the following statements about Uttar Pradesh:

1. Forests constitute about 12.8% of the total geographical area of the State.
2. The districts of Jaunpur (जौनपुर), Ghazipur and Ballia have no forest land.

A. Only 1

B. Both 1 and 2

C. Only 2

D. Neither

Show answer

Ans: C — ISFR 2023 gives **~6.24%** forest cover, not 12.8%. Statement 2 is correct.

8.3 Forest Conservation

Identity: FCA 1980 | Chipko 1973 | NFP 1988 | CAMPA | Godavarman 1996

- **Forest conservation** protects existing forests and restores degraded ones, not just new planting.

- Major threats include deforestation, mining, forest fires, overgrazing, shifting cultivation (**jhum**), and invasive species (आक्रामक विदेशी प्रजाति).
- The **Chipko Movement (चिपको आंदोलन) (1973)** (चिपको आंदोलन) began at **Reni village** in Uttarakhand when villagers hugged trees to stop commercial felling.
- **Sunderlal Bahuguna**, **Chandi Prasad Bhatt** (चंडी प्रसाद भट्ट) (Dasholi Gram Swarajya (स्वराज्य) Mandal, 1964), and **Gaura Devi** (26 March 1974 Reni episode) were key Chipko leaders.
- Major **causes** of deforestation include agriculture, housing, urbanisation, industry, and jhum. **Nagaland** hills becoming barren is classically linked to **shifting cultivation**. **Urbanisation is a cause**, not an "impact/effect," of deforestation — drying springs, biodiversity loss, and soil erosion are impacts.
- Forests are a **renewable** resource under sustainable use. They also absorb CO_2 and release O_2, but that environmental role does **not** alone define renewability (classic A/R trap).
- **Aluminium** is called a **"green metal"** for recyclability/eco-profile — not primarily because it "saves forests by replacing wood."
- **Living root bridges** of trained Ficus roots are characteristic of **Meghalaya**.
- The **Tiwari Committee (1980)** recommended strict control on forest diversion and led to **FCA 1980**.
- **Forest (Conservation) Act, 1980** requires **prior Central Government approval** before forest land is used for non-forest purposes.
- **Compensatory afforestation** must replace diverted forest area, with payment of **net present value (NPV)** into the **CAMPA fund**.
- **CAMPA** was set up in **2009**. The **CAMPA Act, 2016** created national and state fund authorities.
- **National Forest Policy, 1988** replaced the **1952 revenue-oriented policy** and targets **33% forest and tree cover** nationally and **66% in hills and mountains**.
- **NFP 1952** forest classes: **Protected/Conserved Forests, National Forests, Village Forests, Tree Lands** — **National Park is not** a 1952 forest class.
- NFP 1988 includes afforestation of wastelands, reforestation, and wood substitutes / alternate fuels — **not** promoting wide insecticide/pesticide use.
- In **Godavarman v. Union of India (1996)**, the Supreme Court expanded **forest** to include areas meeting the dictionary meaning, not only notified land.
- **Green India Mission (2014)** aims to improve forest quality and carbon sequestration (कार्बन पृथक्करण) under India's climate action plan.
- **Miyawaki** planting creates dense mini-forests quickly on degraded/urban (नगरीय) plots using native species.
- **Article 48A** (DPSP (राज्य के नीति निदेशक तत्व)) directs the State to protect forests. **Article 51A(g)** makes environmental protection a fundamental duty of citizens.
- International Day of Forests = **21 March** (UNGA **2012**). Bhutan constitutionally aims for **≥60%** forest cover.

> **Logic:** FCA = **1980**. WPA = **1972**. NFP current version = **1988**, not 1952. "Urbanisation as deforestation impact" is the classic trap.

8.4 Forest Management

Identity: IFA 1927 | Working plans | FSI/ISFR | Concurrent List (समवर्ती सूची)

- **Forest management** uses scientific planning to harvest forest produce without destroying regeneration capacity.
- The **Indian Forest Act, 1927** empowers states to declare Reserved and Protected Forests and regulate felling.
- A **Working Plan** is the **10-year** scientific management document for each forest division covering inventory, regeneration, and felling schedules.
- **Sustainable yield** means harvest must not exceed the forest's regeneration capacity.
- **Forest Survey of India (FSI) (भारतीय वन सर्वेक्षण), Dehradun** publishes the biennial **India State of Forest Report (ISFR)**.
- **Forest Research Institute (FRI) (वन अनुसंधान संस्थान), Dehradun** (roots in **1906** Imperial FRI) is a premier forestry research/training institute and deemed university.
- **Indian Council of Forestry Research and Education (ICFRE), Dehradun** coordinates forestry research and revised type classification work.
- **Indian Institute of Forest Management (IIFM)** is at **Bhopal**. **Survey of India** is a mapping agency under DST — **not** an environment body.
- **Recorded Forest Area** is land legally notified as forest, about **23%** of India's geographical area.
- About **27%** of India's forest cover lies **outside** Recorded Forest Area according to ISFR 2023.
- **Forests** are on the **Concurrent List** (समवर्ती सूची), so both Centre and states can legislate.
- **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) sets national policy. State Forest Departments implement protection, working plans, and JFM on the ground.
- **Van Panchayats** in Uttarakhand manage local forest patches through customary community institutions.

> **Logic:** ISFR is published by **FSI**, not CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड), ICFRE, or MoEFCC directly.

8.5 Afforestation

Identity: New forest on degraded/non-forest land | NAP | GIM | compensatory planting

- **Afforestation** creates forest cover on land that had no forest or lost canopy below the 10% threshold.
- **Reforestation** replants harvested forest. **Restoration** rebuilds full ecological function.
- **Compensatory afforestation** under FCA is **mandatory** when notified forest land is diverted.
- The developer pays **net present value (NPV)** of forest services into the **CAMPA fund**.
- **National Afforestation Programme (NAP)** was launched in **2000** through **Forest Development Agencies (FDAs)**.
- **Green India Mission (2014)** targets improvement of **5 million hectares** of forest and tree cover.
- **Nagar Van Scheme (2020)** promotes urban forests in cities.

- The **Miyawaki method** creates dense urban mini-forests with native species planted close together on degraded or urban land.
- **National Afforestation and Eco-Development Board (NAEB)** was set up in **1992**.
- Bamboo grown on **non-forest** land was omitted from the definition of "tree" under the **Indian Forest (Amendment) Act, 2017** — freer harvest/transport outside forests; FRA still treats bamboo as **minor forest produce**.
- **Native species** planting is preferred under NFP 1988 and Green India Mission over exotic monocultures such as eucalyptus.
- Farm and agroforestry plantations add to **tree cover** and **Trees Outside Forests (TOF)** in ISFR, which helps plain states like UP.

> **Logic:** CAMPA money is used mainly for afforestation and forest protection, not wildlife alone.

8.6 Social Forestry

Identity: NCA 1976 | Non-forest land | Fuelwood–fodder focus

- **Social forestry** grows trees on **non-forest land** to meet local fuelwood, fodder, and timber needs.
- The **National Commission on Agriculture (1976)** promoted it to address India's **fuelwood famine**.
- **Farm forestry** is trees planted by farmers on private land, such as poplar in western UP for plywood markets.
- **Community forestry** uses village commons and panchayat land for fuelwood and fodder.
- **Extension forestry** lines roadsides, canals, and institutional land with trees.
- **NFP 1988** endorsed social forestry as part of the **33% cover target** and stressed women's participation.
- Social forestry **reduces pressure** on natural forests by supplying needs locally.
- **Kerala** was an early successful social forestry state in the 1970s–80s.
- **Social forestry is not JFM**. JFM co-manages existing degraded forest, not non-forest land.
- **Social forestry is not agroforestry**. Agroforestry integrates trees with active crop or livestock systems on farmland.

> **Logic:** Social forestry roots lie in **NCA 1976**, not JFM 1990.

8.7 Agroforestry

Identity: National Policy 2014 | ICAR-CAFRI Jhansi | Trees + crops/livestock

- **Agroforestry** deliberately combines trees with crops and/or livestock on the same land unit.
- The **National Agroforestry Policy, 2014** was India's first comprehensive agroforestry policy.
- **ICAR-CAFRI (Central Agroforestry Research Institute)** is located in **Jhansi, Uttar Pradesh**.
- **ICRAF (World Agroforestry Centre)** is headquartered in **Nairobi**, not India.

- Common systems include **silvopasture**, **alley cropping**, **home gardens**, and **windbreaks**.
- **Multi-Purpose Tree Species (MPTS)** such as poplar, eucalyptus, teak, and bamboo are chosen for market value and fast growth.
- Agroforestry improves soil fertility, diversifies farmer income, and adds to **tree cover** in ISFR.
- **Taungya** allows temporary farming between rows of planted trees in forest department plantations.
- FAO treats agroforestry as a **climate-smart agriculture** practice.

> **Logic:** Agroforestry Policy = **2014**, not 2006. CAFRI = **Jhansi** (झांसी), not Banda.

PYQ — UPPCS Prelims 2021, Q15

Which is NOT correctly matched? A. ICRAF — Nairobi B. IIFM — Bhopal C. CAFRI — Banda D. TERI — New Delhi

Show answer

Ans: C — Central Agroforestry Research Institute is in **Jhansi, UP**.

8.8 Joint Forest Management (JFM)

Identity: MoEF resolution 1990 | VSS/FPC | 50% benefit share

- **Joint Forest Management (JFM)** is a partnership between the State Forest Department and village communities.
- JFM was launched through a **MoEF resolution in June 1990**.
- It aims to **regenerate degraded forest** with community protection and shared benefits.
- Village bodies called **Van Samrakshan Samiti (समिति) (VSS)** or **Forest Protection Committee (FPC)** implement JFM.
- Communities receive about **50% of net income** from regenerated forest produce after final felling.
- **Intermediate produce** such as grass, fodder, and fallen wood is often shared fully with the village during protection.
- **Micro-plans** guide silviculture, protection, and livelihood activities for each JFM unit.
- **NFP 1988** provided the policy foundation for community participation that JFM operationalized.
- India has over **100,000 JFM committees**, one of the world's largest participatory forestry programmes.
- **JFM is an executive scheme**, not a statute like FRA 2006.
- **JFM does not grant permanent land title**. FRA can grant recognized forest rights.

> **Logic:** UPPCS 2021 asked JFM launch year — answer **1990**.

8.9 Forest Rights Act

Identity: FRA 2006 | ST + OTFD | Gram Sabha | IFR 4 ha cap

- The **Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006** corrects historical injustice from colonial forest laws.
- **Rules were notified in 2008**.
- It covers **Scheduled Tribes** and **Other Traditional Forest Dwellers (OTFDs)** who depend on forests.
- An **OTFD** must have depended on forest land for livelihood for at least **three generations** prior to **13 December 2005**.
- **Individual Forest Rights (IFR)** allow habitation and cultivation on forest land up to **4 hectares** per household.
- Claimants must have occupied the land as of **13 December 2005**.
- **Community Forest Rights (CFR)** cover grazing, NTFP collection, nistar, and community forest management.
- **Community Forest Resource (CFR) (सामुदायिक वन संसाधन अधिकार) management rights** allow a village to protect and manage its traditional forest boundary.
- The **Gram Sabha** (ग्राम सभा) receives, verifies, and passes resolutions on forest rights claims.
- Claims then go to **SDLC (Sub-Divisional Level Committee)** and **DLC (District Level Committee)** for final approval.
- Rights are **heritable** but **not transferable or alienable**. Titles are registered in both spouses' names.
- **No eviction** of eligible claimants is allowed until the recognition process is complete.
- **Critical Wildlife Habitat (वास स्थान)** provisions allow relocation from protected areas only after strict scientific and consent procedures.
- In **Niyamgiri (2013)**, Gram Sabhas rejected Vedanta (उत्तर मीमांसा) bauxite mining under FRA community consent powers.
- **FRA grants rights**. **FCA controls diversion**. Both can apply to the same forest landscape.

> **Logic:** IFR cap = **4 ha**, not 5 or 10. Cutoff date = **13 December 2005**.